

Cattle Breed Classification through Multiple Approaches

A thesis submitted in the fulfilment of the requirements for the degree of

BACHELOR OF SCIENCE in COMPUTER SCIENCE & ENGINEERING

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A blue arrow pointing to the right, with a white outline and a slight 3D effect.

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PROBLEM DOMAIN

- Detect and identify the cattle breed.
- Classify the cattle breed.



(a) White Native



(b) Shahiwal



(c) Holstein Friesian



(d) Black Angus



(e) Red Chittagong



OBJECTIVE

- To contribute to advancements in agricultural technology by providing efficient and accurate tools for cattle classification, aiding farmers and researchers in optimizing livestock practices.

MOTIVATIONS

- Replace ancient cattle detection technique with automation
- Provide the farmers an effective system to identify cattle breed by using captured images
- Overcome limitations of existing methods
 - Small dataset
 - Overfitting
 - Lower accuracy



APPLICATIONS



- Automated systems for farmers.
- Facilitate the decision-making process for livestock management.

RELATED WORKS

Table 1. Related works in Cattle breed recognition

No.	Author's name	Datasetsize	Data Augmentation	Analysis Epochs	Model	Accuracy
1	Jane A. Researcher et al.	Large	Yes	50	Convolutional Neural Network (CNN)	92%
2	Dr. Mark Scientist et al.	8,000	Not Specified	Not Specified	Support Vector Machines(SVM), Random Forests, Deep Neural Networks	85% (SVM)
3	Ahmed Investigator et al.	12,000	Yes	50	Deep Learning Model	88%

PLATFORMS

- **Operating System**

1. Windows 10

- **IDE**

1. Google Colab

- **Tools Requirement**

1. Python
2. TensorFlow and Keras
3. scikit-learn
4. OpenCV
5. Matplotlib
6. PIL
7. NumPy
8. Seaborn
9. mlxtend

BACKGROUND STUDY



Categories of Cattle Breed:

- Holstein Friesian



Figure 1: Holstein Images

Characteristics:

1. Holstein cows have distinctive black and white or red and white markings.
2. An adult cow can weigh between 690-770 kilograms and measure a height of about 140-165 cm.
3. The Holstein breed, originating almost 2000 years ago in the Netherlands, emerged from the crossbreeding of black Batavian animals in Germany and white Friesian cattle in Holland.

BACKGROUND STUDY (CONT.)

Categories of Cattle Breed:

- Black Angus



Figure 2: Black Angus Images

Characteristics:

1. Black Angus cattle are naturally polled and can be black or red in colour although black is the dominant color, white may occasionally appear on the udder
2. Adult Black Angus usually weigh between 450 to 1,000 kg and stand at a shoulder height ranging from 4.5 to 6 feet.
3. Angus, cattle breed has a deep-rooted history and originated in Scotland, particularly in the counties of Aberdeen and Angus.

BACKGROUND STUDY (CONT.)

Categories of Cattle Breed:

- Red Chittagong



Figure 3: Red Chittagong Images

Characteristics:

1. Red Chittagong cattle are reddish in color. Red Chittagong, also known as RCC.
2. Mature bulls weigh around 342 kg, while cows weigh approximately 180 kg.
3. The Red Chittagong breed, has its origins in the Chittagong Hill Tracts region of Bangladesh.

BACKGROUND STUDY (CONT.)

Categories of Cattle Breed:

- Shahiwal



Figure 4: Shahiwal Images

Characteristics:

1. Their colour can range from reddish brown through to the more predominant red, with varying amounts of white on the neck, and the underline.
2. Average body weight of the mature bulls is between 400 and 500 kg.
3. Average body weight of the mature cows is between 700 and 800 kg.
4. The Sahiwal, an ancient cattle breed originating from the region that is now part of Pakistan, has a rich history dating back to the early 20th century.

BACKGROUND STUDY (CONT.)

Categories of Cattle Breed:

- White Native



Figure 5: White Native Images

Characteristics:

1. The Cattle colors can vary widely, including shades of brown, black, white, and a mix of these colors.
2. The White Native Cattle breed is relatively small.
3. Adult bulls weighing approximately 400 kg and mature cows weighing around 300 kg.
4. The Native Cattle, colloquially known as "Kalabaw," have a deep-rooted history in the Philippines, reflecting the nation's rich agricultural heritage.

DATASET

Original Dataset

Total 2050 cattle breed images

- ❑ Holstein Friesian (558 images)
- ❑ Black Angus (330 images)
- ❑ Red Chittagong (470 images)
- ❑ Shahiwal (380 images)
- ❑ White Native (312 images)

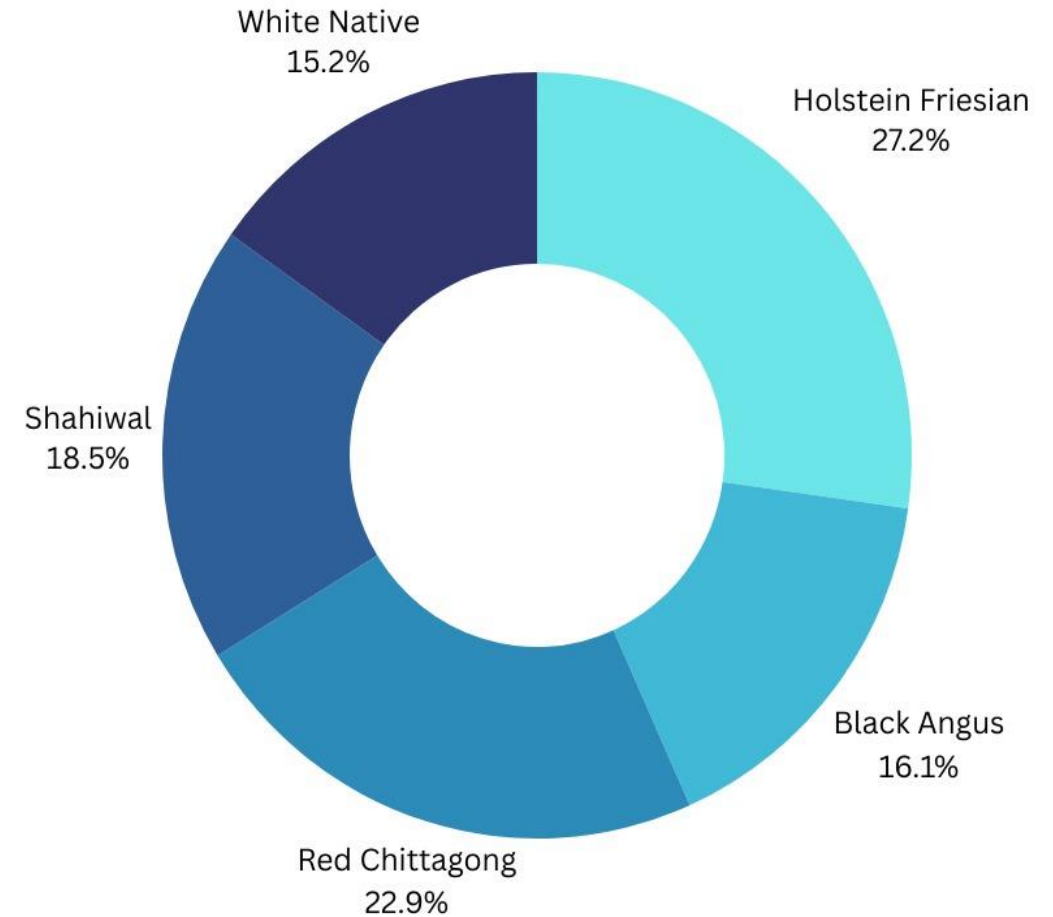


Figure 6: Original Dataset Distribution

DATASET: Augmented

Dataset after augmentation process

Total 5025 cattle breed images

- ☐ Holstein Friesian (1000 images)
- ☐ Black Angus (1025 images)
- ☐ Red Chittagong (1000 images)
- ☐ Shahiwal (1000 images)
- ☐ White Native (1000 images)

Cattle Breed Classification

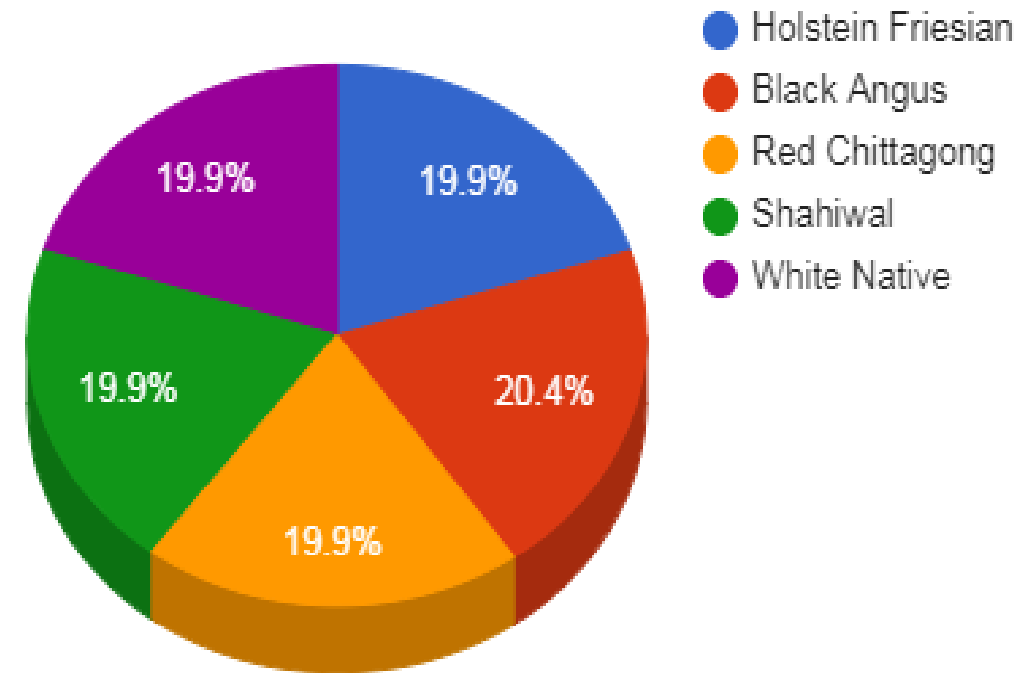


Figure 7: Dataset after Augmentation Distribution



Table 2. Dataset description of Cattle breed classification

	Names	Original images	Distribution Techniques	
			No. of training images (70%)	No. of test images (30%)
Cattle breeds	Holstein Friesian	1000	710	290
	Black Angus	1025	706	319
	Red Chittagong	1000	714	286
	Shahiwal	1000	682	318
	White Native	1000	706	294

METHODOLOGY

Overall Working Process of Cattle Breed Classification

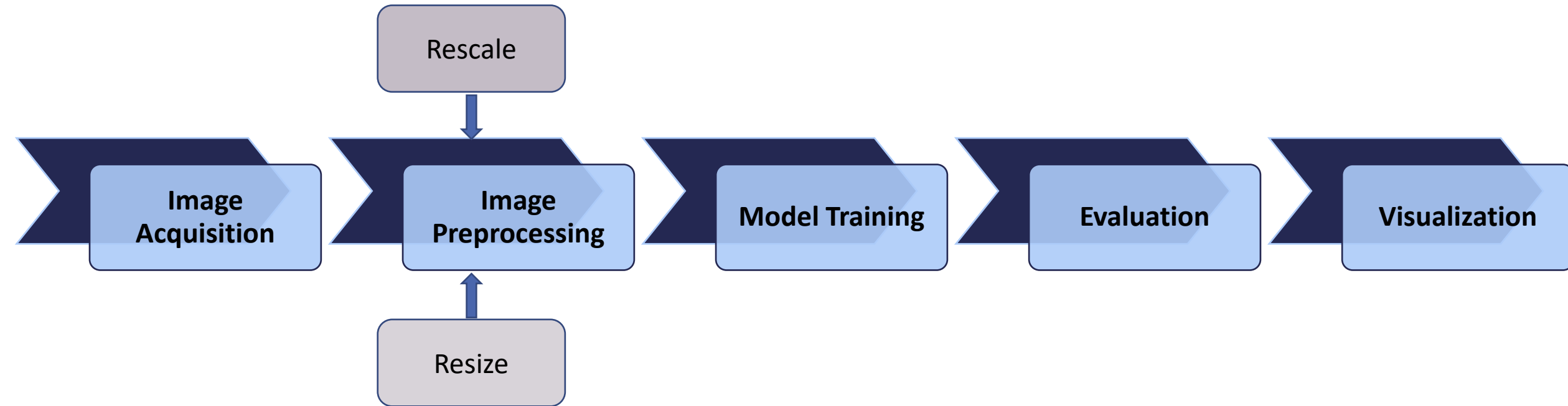


Figure 8: Overall Working Process of Cattle Breed Classification

METHODOLOGY (CONT.)

Image Preprocessing

- ❑ Resizing images into 300*300



(a)



(b)

Figure 9: (a) Red Chittagong (X*Y) (b) Red Chittagong (300*300)

METHODOLOGY (CONT.)

Data Augmentation

- Rescale
- Rotation
- Shift
- Shear
- Flip
- Zoom

METHODOLOGY (CONT.)

Work Flow of Data Preprocessing

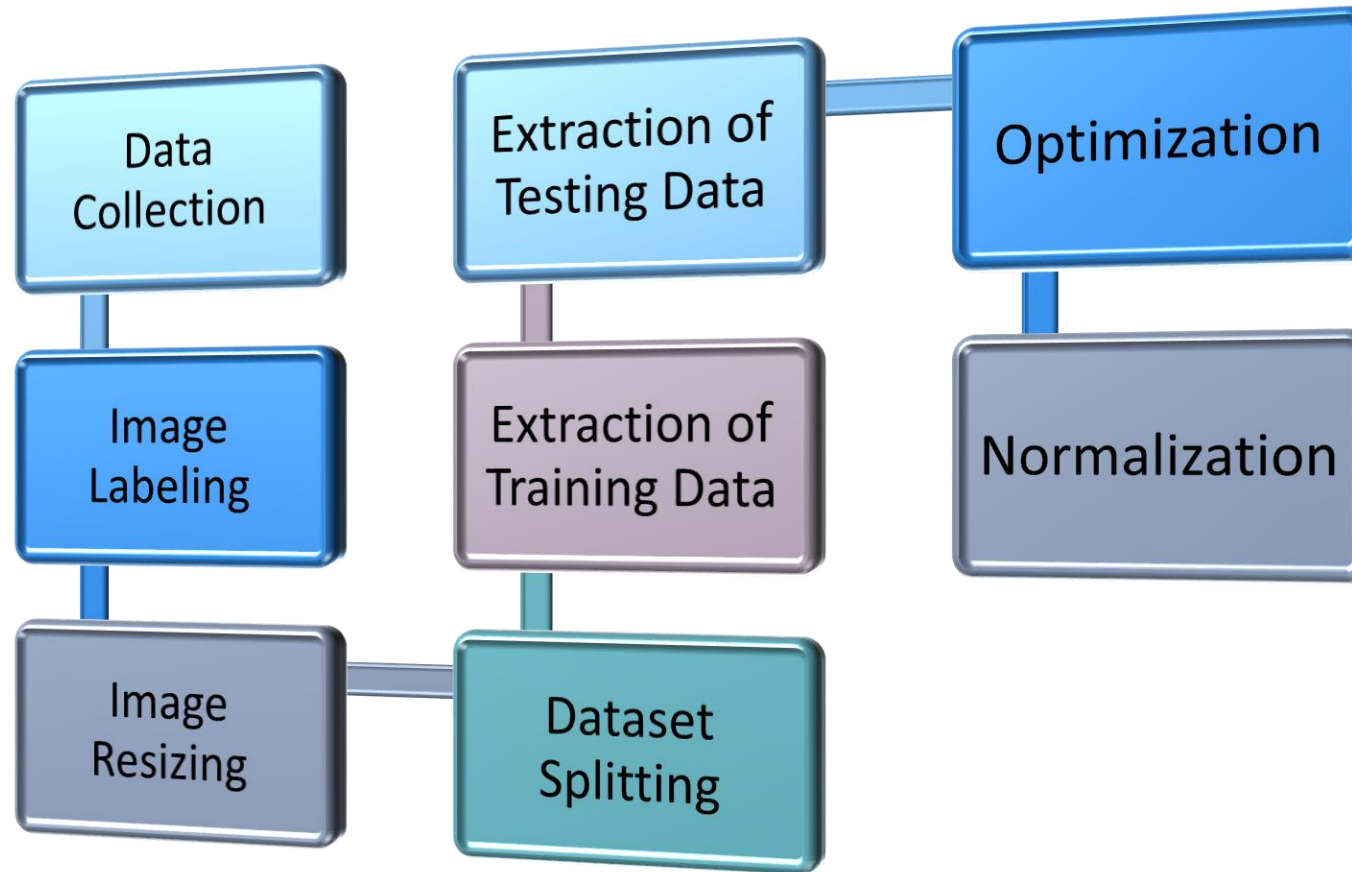


Figure 10: Work Flow of data preprocessing

METHODOLOGY (CONT.)

Working Procedure of our CNN model

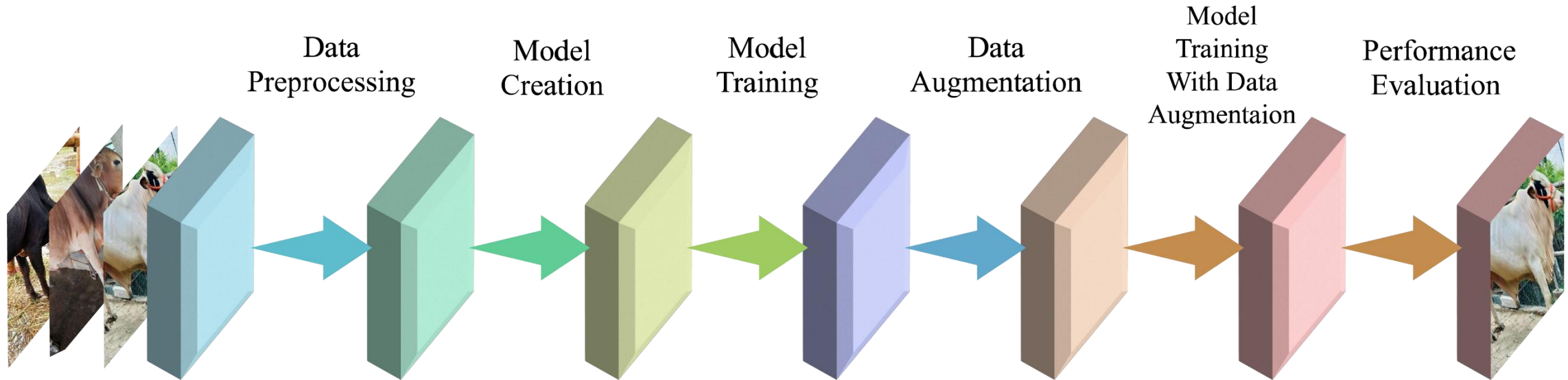


Figure 11: CNN Work Flow

METHODOLOGY (CONT.)

Working Procedure of our SVM model

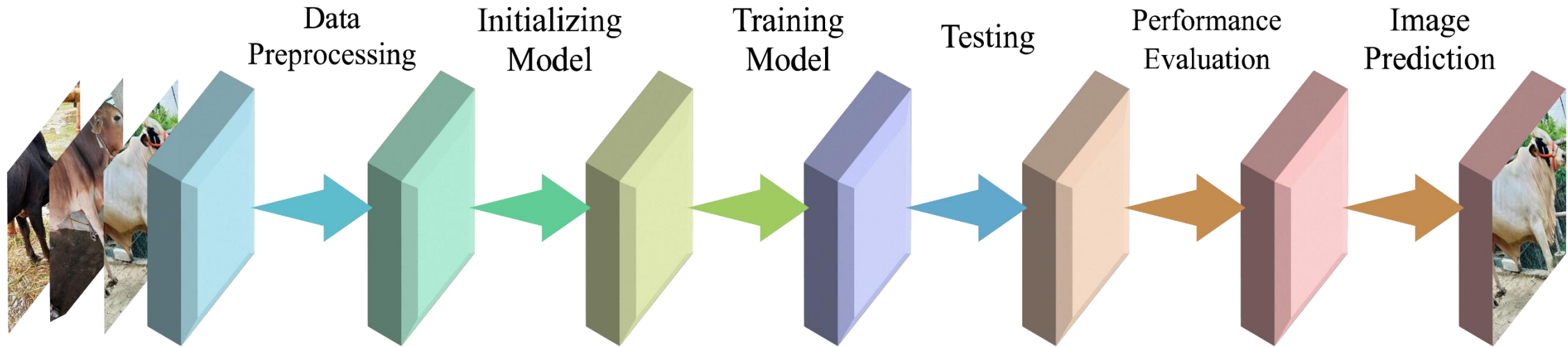


Figure 12: SVM Work Flow

METHODOLOGY (CONT.)

Working Procedure of our Decision Tree model

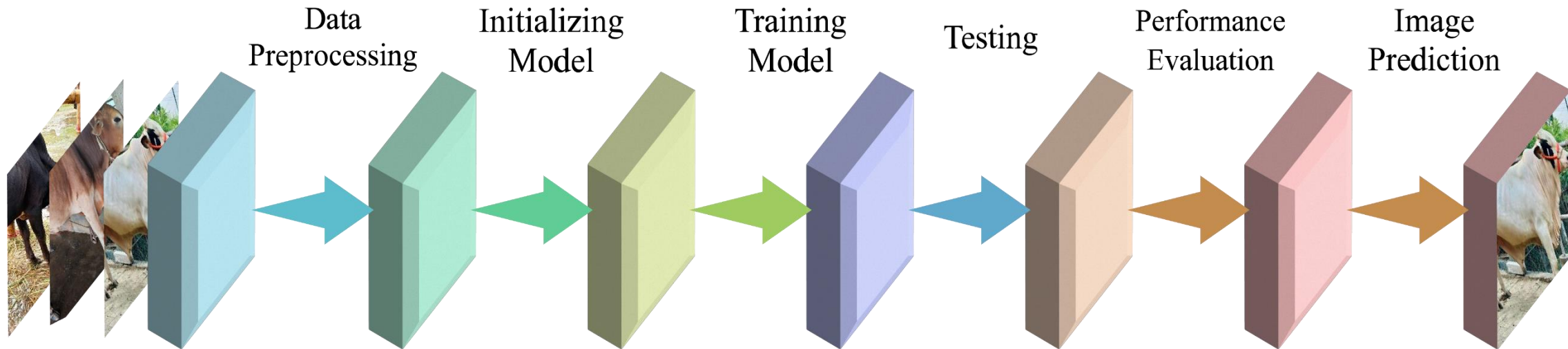


Figure 13: Decision Tree Work Flow

METHODOLOGY (CONT.)

Working Procedure of our KNN model

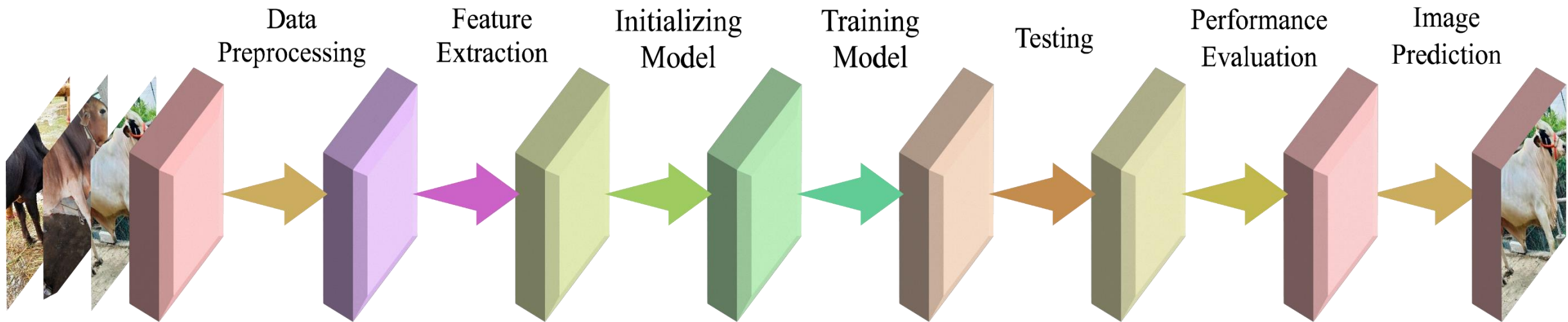


Figure 14: KNN Work Flow

METHODOLOGY (CONT.)

Working Procedure of our Random Forest model

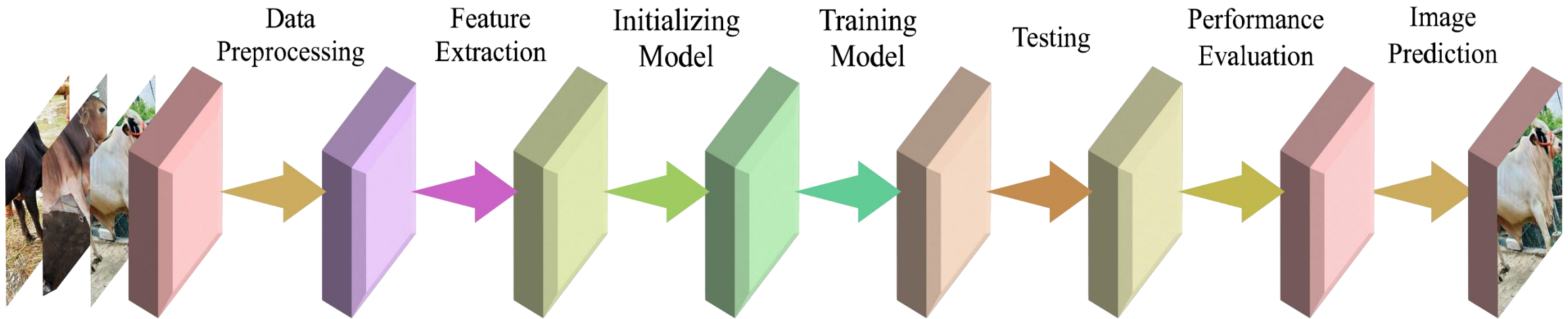


Figure 15: Random Forest Work Flow

RESULT & OBSERVATION

Training accuracy vs Testing accuracy and Training loss vs Testing loss of CNN Model

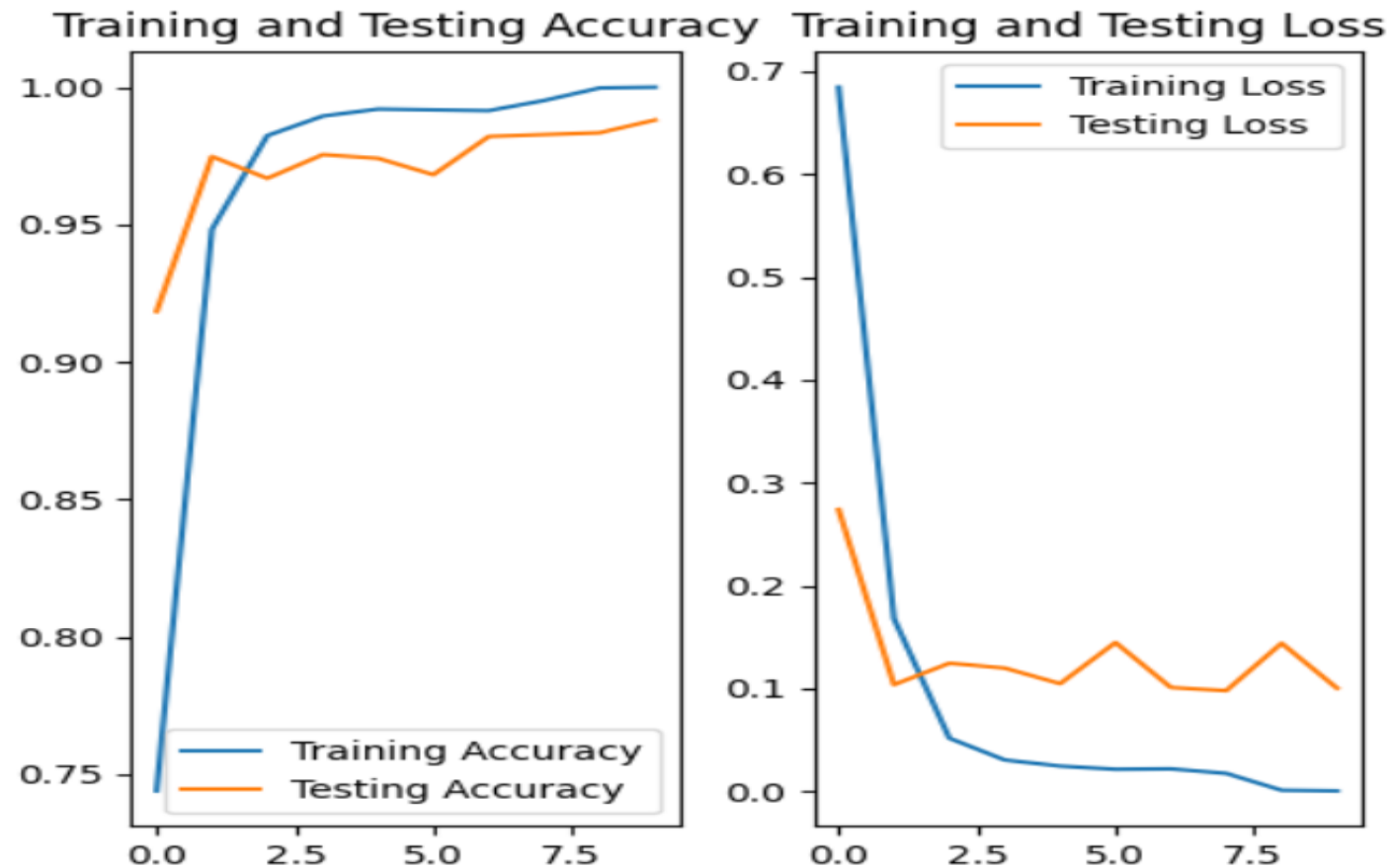
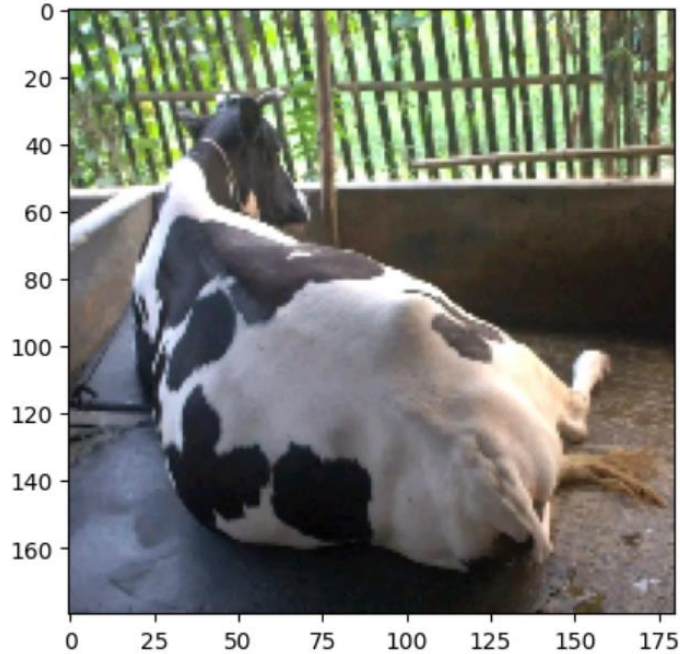


Figure 16: Evaluation of Accuracy and loss with The Number of Epochs

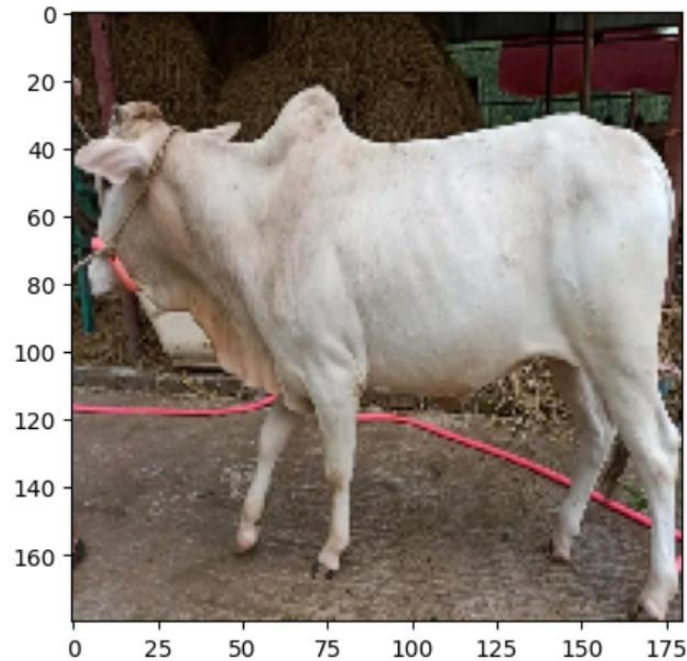
RESULT & OBSERVATION (CONT.)

Single prediction of cattle breed

This image belongs to Holstein Friesian and Accuracy 99.54 %.



This image belongs to White Native and Accuracy 99.71 %.



This image belongs to Black Angus and Accuracy 99.75 %.

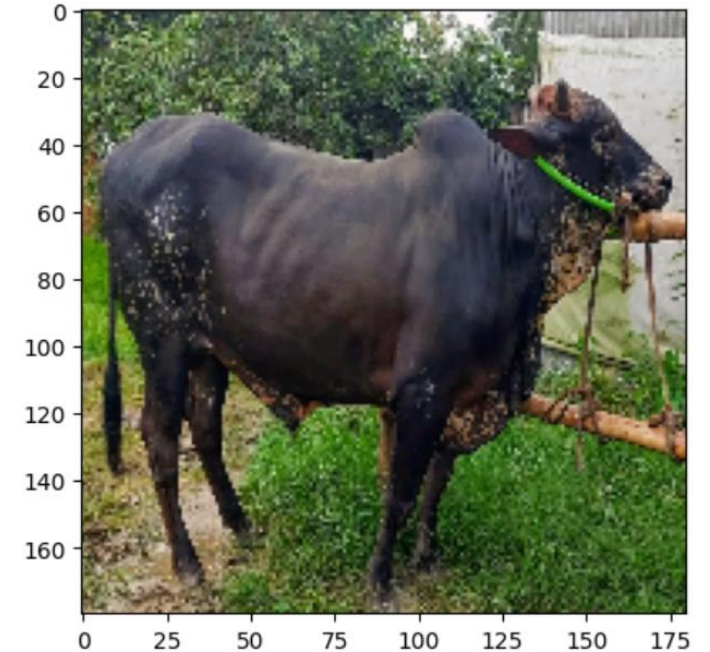


Figure 17: Correctly classified Cattle breeds

RESULT & OBSERVATION (CONT.)

Single prediction of cattle(cont.)



The predicted class for the unknown image is: Red Chittagong



The predicted class for the unknown image is: Black Angus

Figure 18: Misclassified Cattle breeds

RESULT & OBSERVATION (CONT.)

Confusion matrix of CNN

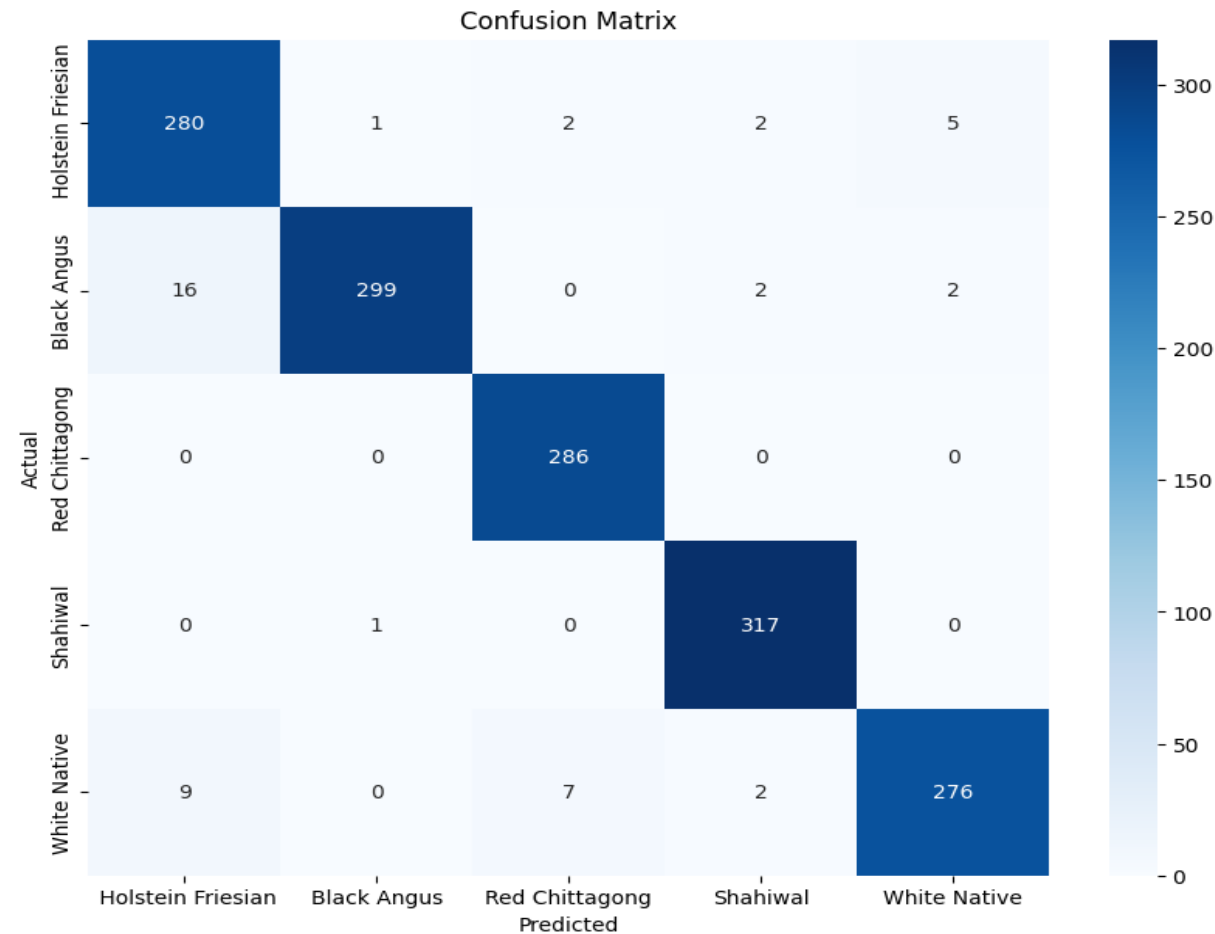


Figure 19: Confusion Matrix for recognizing cattle breed using CNN

RESULT & OBSERVATION (CONT.)

Classification Report of CNN

Classification Report:

	precision	recall	f1-score	support
Holstein Friesian	0.92	0.97	0.94	290
Black Angus	0.99	0.94	0.96	319
Red Chittagong	0.97	1.00	0.98	286
Shahiwal	0.98	1.00	0.99	318
White Native	0.98	0.94	0.96	294
accuracy			0.97	1507
macro avg	0.97	0.97	0.97	1507
weighted avg	0.97	0.97	0.97	1507

Figure 20: Classification Report of CNN

RESULT & OBSERVATION (CONT.)

Confusion matrix of SVM

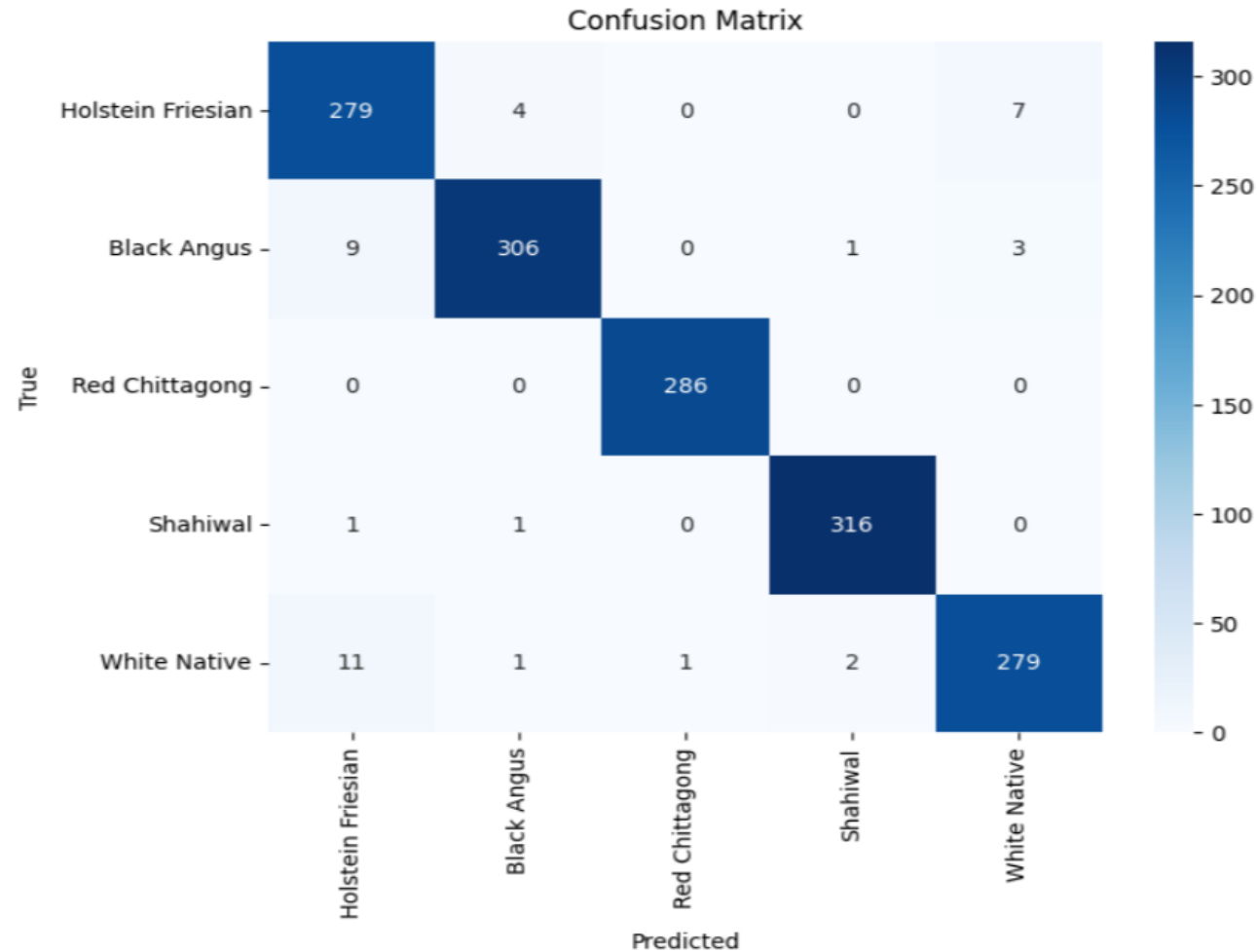


Figure 21: Confusion Matrix for recognizing cattle breed using SVM

RESULT & OBSERVATION (CONT.)

Classification Report of SVM

Classification Report:

	precision	recall	f1-score	support
Holstein Friesian	0.93	0.96	0.95	290
Black Angus	0.98	0.96	0.97	319
Red Chittagong	1.00	1.00	1.00	286
Shahiwal	0.99	0.99	0.99	318
White Native	0.97	0.95	0.96	294
accuracy			0.97	1507
macro avg	0.97	0.97	0.97	1507
weighted avg	0.97	0.97	0.97	1507

Figure 22: Classification Report of SVM

RESULT & OBSERVATION (CONT.)

Confusion matrix of Decision Tree

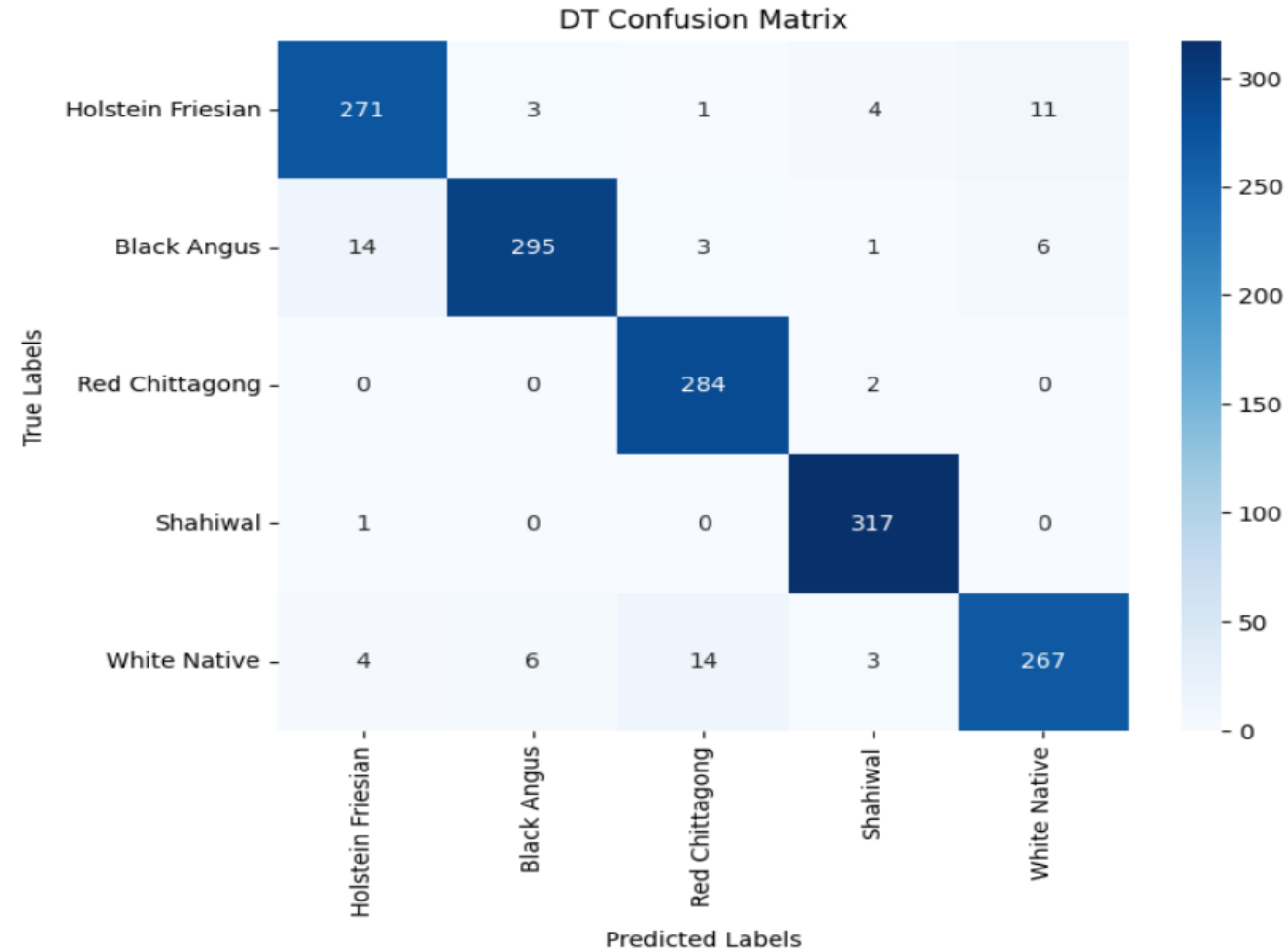


Figure 23: Confusion Matrix for recognizing cattle breed using Decision Tree

RESULT & OBSERVATION (CONT.)

Classification Report of Decision Tree

DT Classification Report:

	precision	recall	f1-score	support
Holstein Friesian	0.93	0.93	0.93	290
Black Angus	0.97	0.92	0.95	319
Red Chittagong	0.94	0.99	0.97	286
Shahiwal	0.97	1.00	0.98	318
White Native	0.94	0.91	0.92	294
accuracy			0.95	1507
macro avg	0.95	0.95	0.95	1507
weighted avg	0.95	0.95	0.95	1507

Figure 24: Classification Report of Decision Tree



RESULT & OBSERVATION (CONT.)



Confusion matrix of KNN

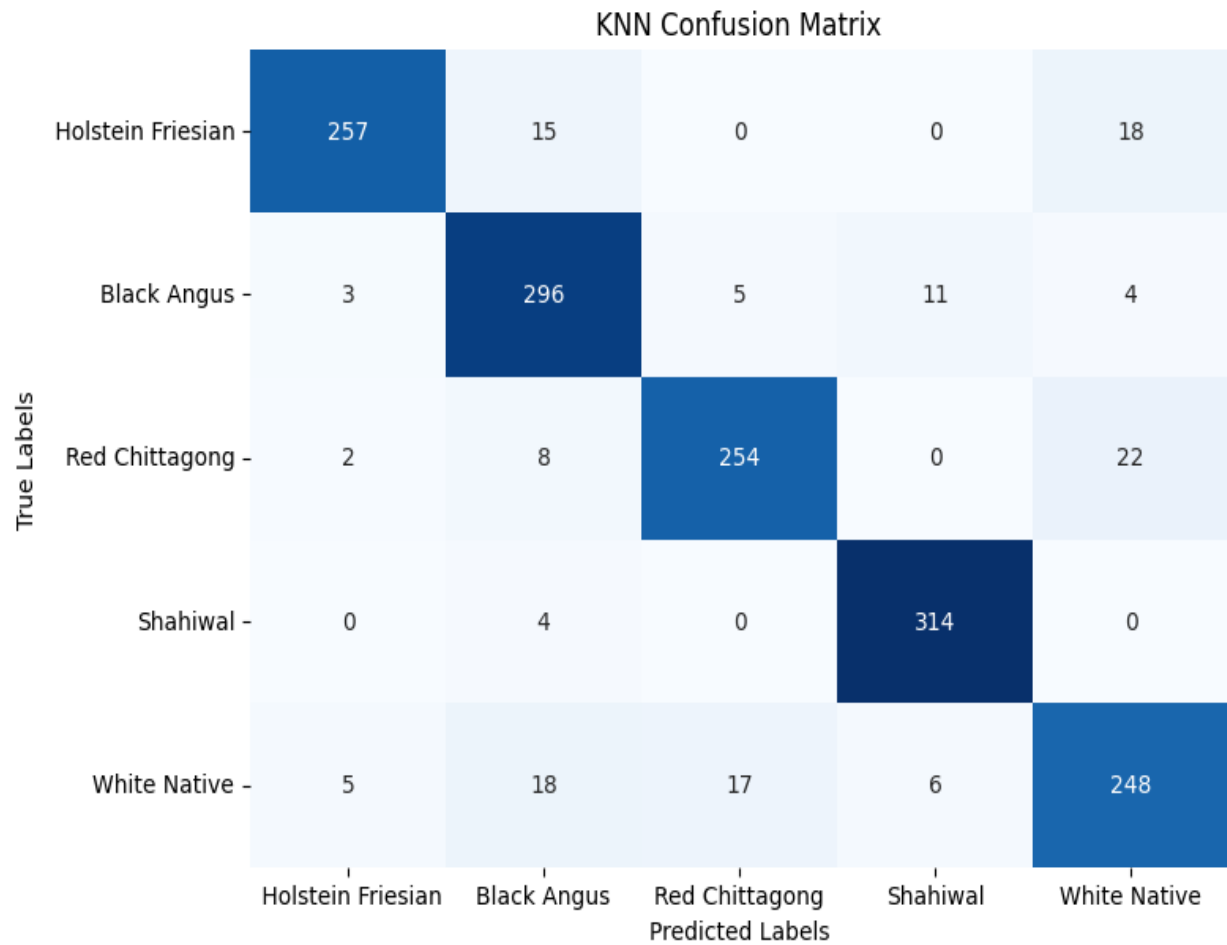


Figure 25: Confusion Matrix for recognizing cattle breed using KNN

RESULT & OBSERVATION (CONT.)

Classification Report of KNN

KNN Classification Report:

	precision	recall	f1-score	support
Holstein Friesian	0.96	0.89	0.92	290
Black Angus	0.87	0.93	0.90	319
Red Chittagong	0.92	0.89	0.90	286
Shahiwal	0.95	0.99	0.97	318
White Native	0.85	0.84	0.85	294
accuracy			0.91	1507
macro avg	0.91	0.91	0.91	1507
weighted avg	0.91	0.91	0.91	1507

Figure 26: Classification Report of KNN

RESULT & OBSERVATION (CONT.)

Confusion matrix of Random Forest

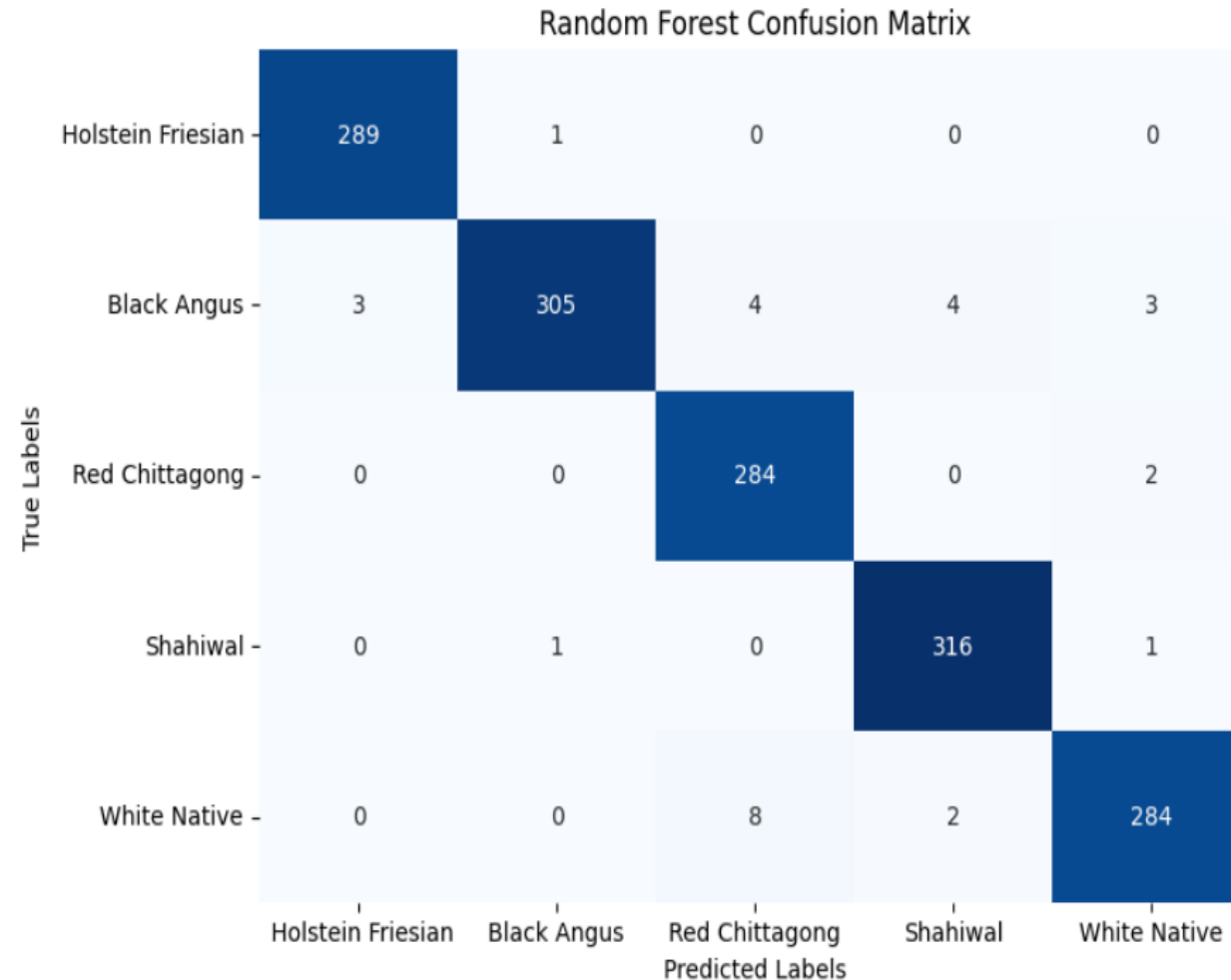


Figure 27: Confusion Matrix for recognizing cattle breed using Random

RESULT & OBSERVATION (CONT.)

Classification Report of Random Forest

Random Forest Classification Report:

	precision	recall	f1-score	support
Holstein Friesian	0.99	1.00	0.99	290
Black Angus	0.99	0.96	0.97	319
Red Chittagong	0.96	0.99	0.98	286
Shahiwal	0.98	0.99	0.99	318
White Native	0.98	0.97	0.97	294
accuracy			0.98	1507
macro avg	0.98	0.98	0.98	1507
weighted avg	0.98	0.98	0.98	1507

Figure 28: Classification Report of Random Forest



RESULT & OBSERVATION (CONT.)



Table 3: Overall Result of Cattle Breed Classification

Model	precision	recall	f1-score	Accuracy
Support Vector Machine	0.9731	0.9728	0.9728	97%
Decision Tree	0.9517	0.9516	0.9513	95%
K-nearest Neighbor	0.9095	0.9084	0.9083	91%
Random Forest	0.9810	0.9808	0.9807	98%
CNN	0.9675	0.9677	0.9672	97%

RESULT & OBSERVATION (CONT.)

Comparison of each model

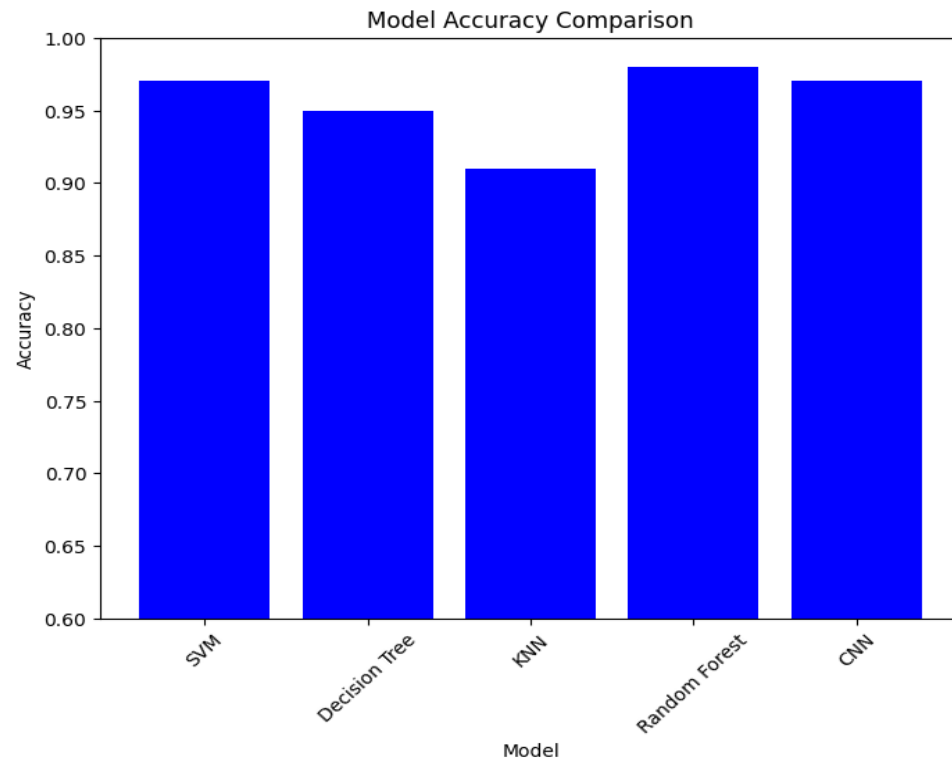


Figure 29: Model Accuracy Comparison

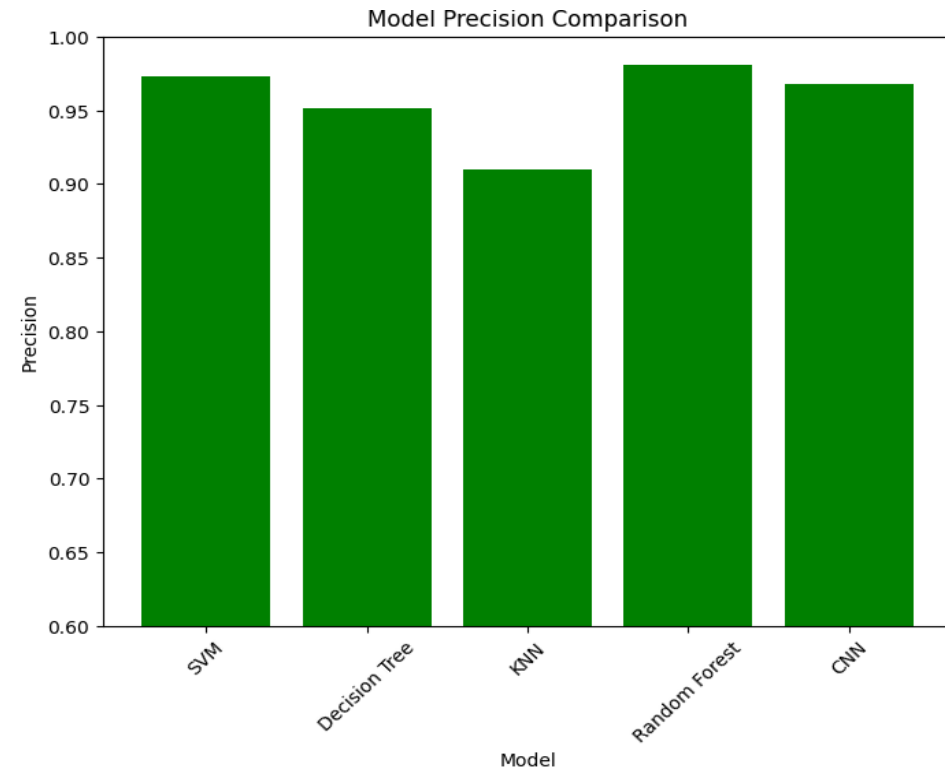


Figure 30: Model Precision Comparison

RESULT & OBSERVATION (CONT.)

Comparison of each model

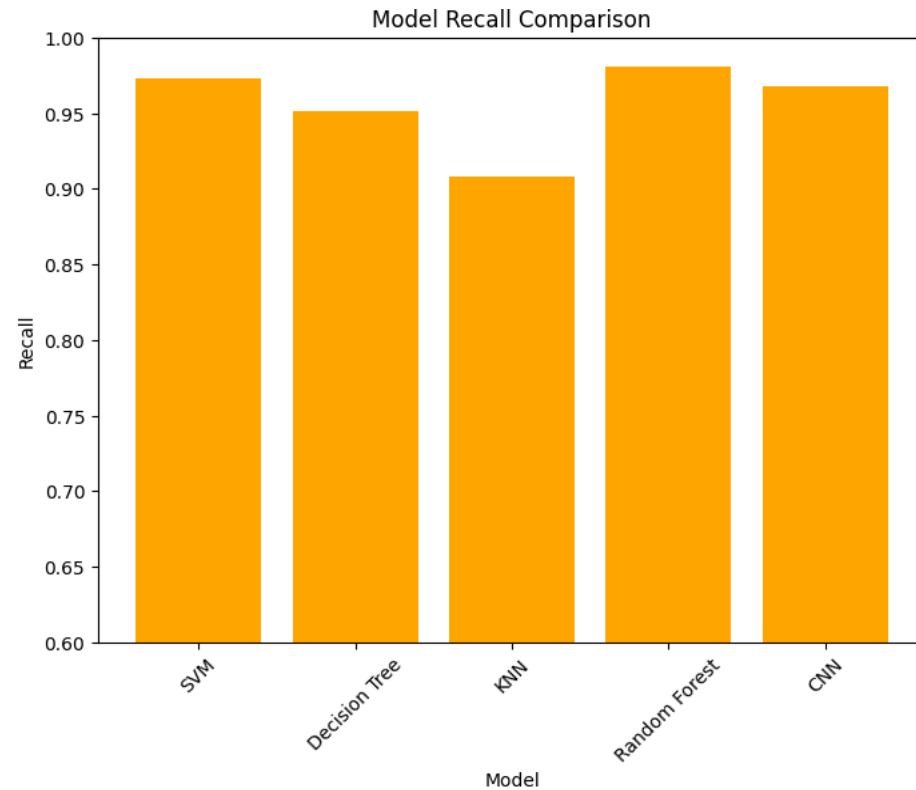


Figure 31: Model Recall Comparison

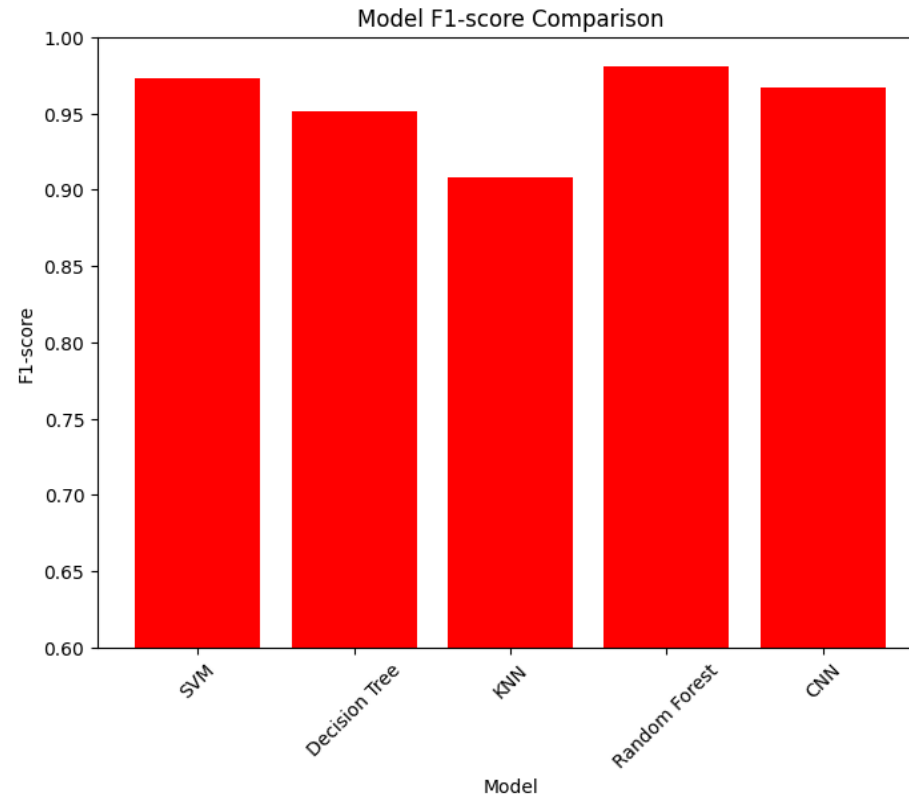


Figure 32: Model F1-score Comparison

RESULT & OBSERVATION (CONT.)

Measuring performance according to ROC curve for CNN model

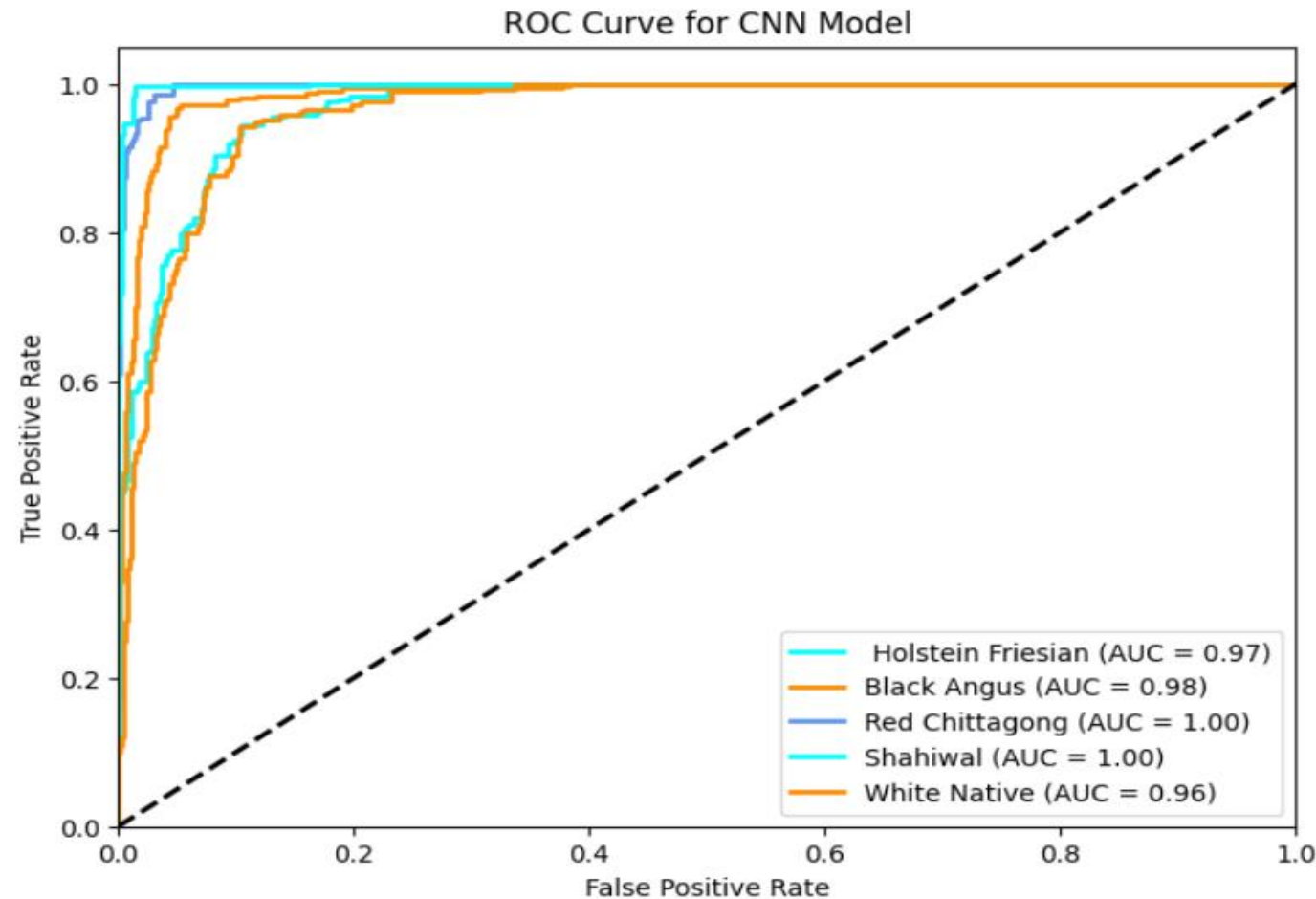


Figure 33: ROC curve of cattle breed recognition

RESULT & OBSERVATION (CONT.)

Measuring performance according to ROC curve for SVM model

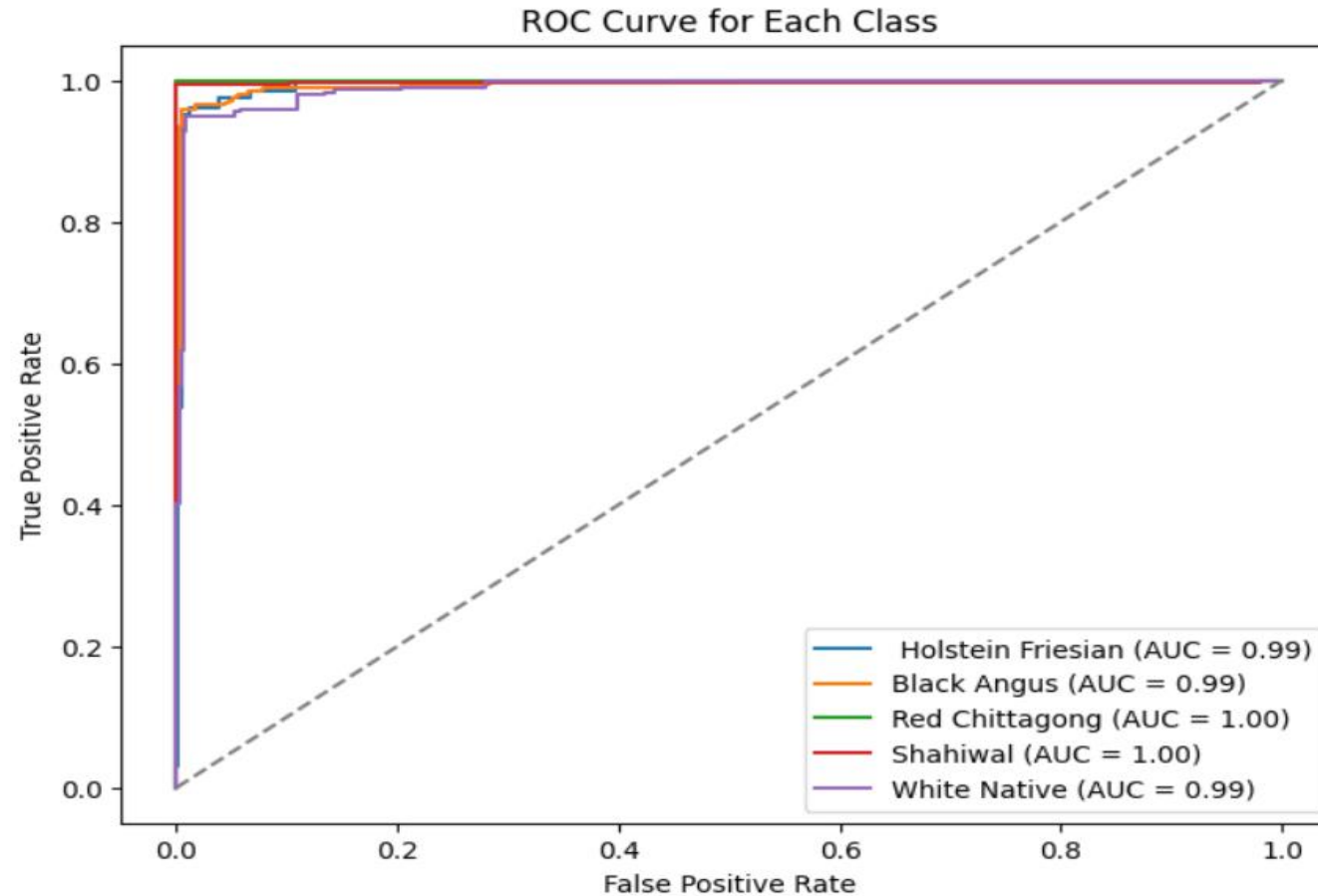


Figure 34: ROC curve of cattle breed recognition



RESULT & OBSERVATION (CONT.)



Measuring performance according to ROC curve for Decision Tree

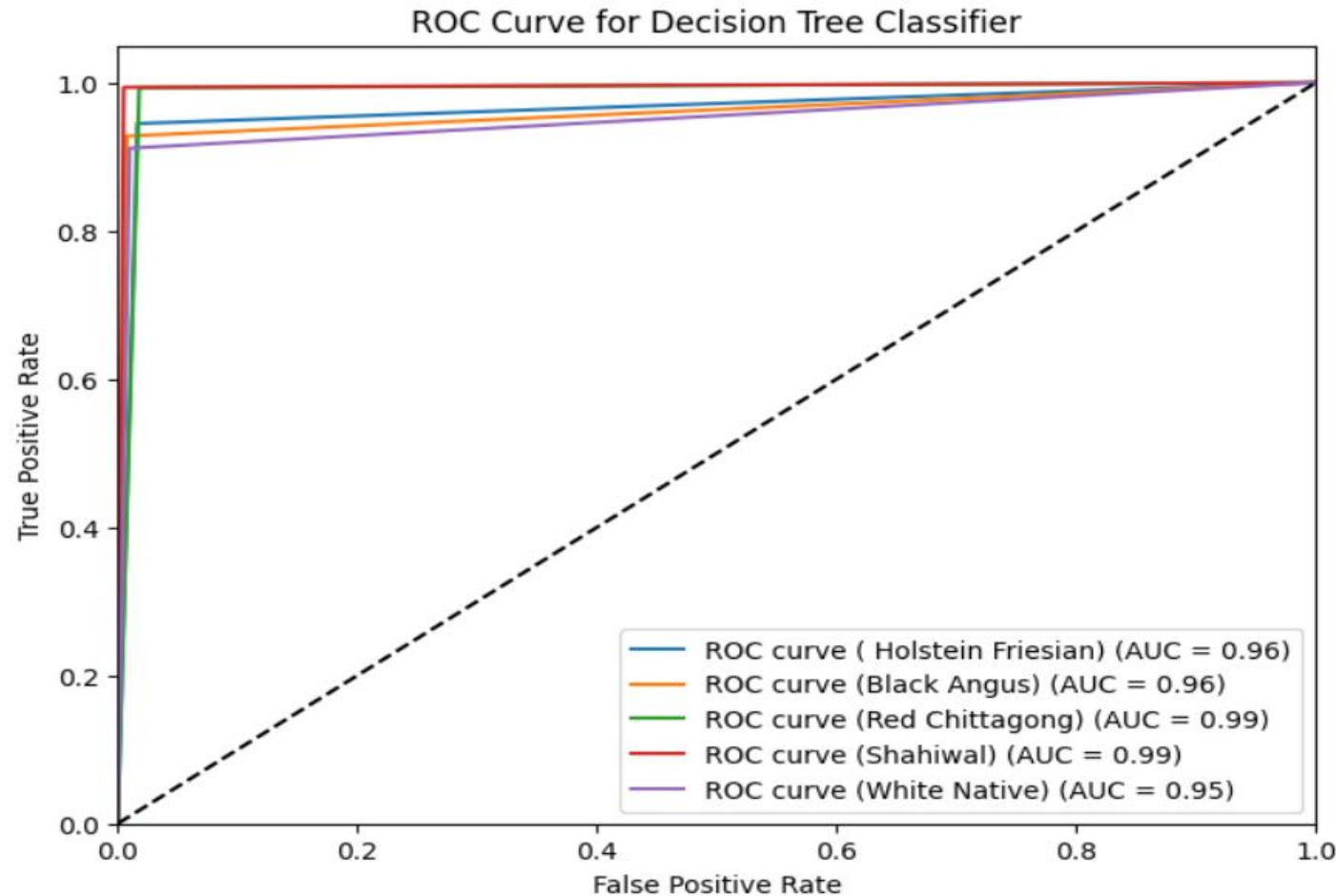


Figure 35: ROC curve of cattle breed recognition

RESULT & OBSERVATION (CONT.)

Measuring performance according to ROC curve for KNN model

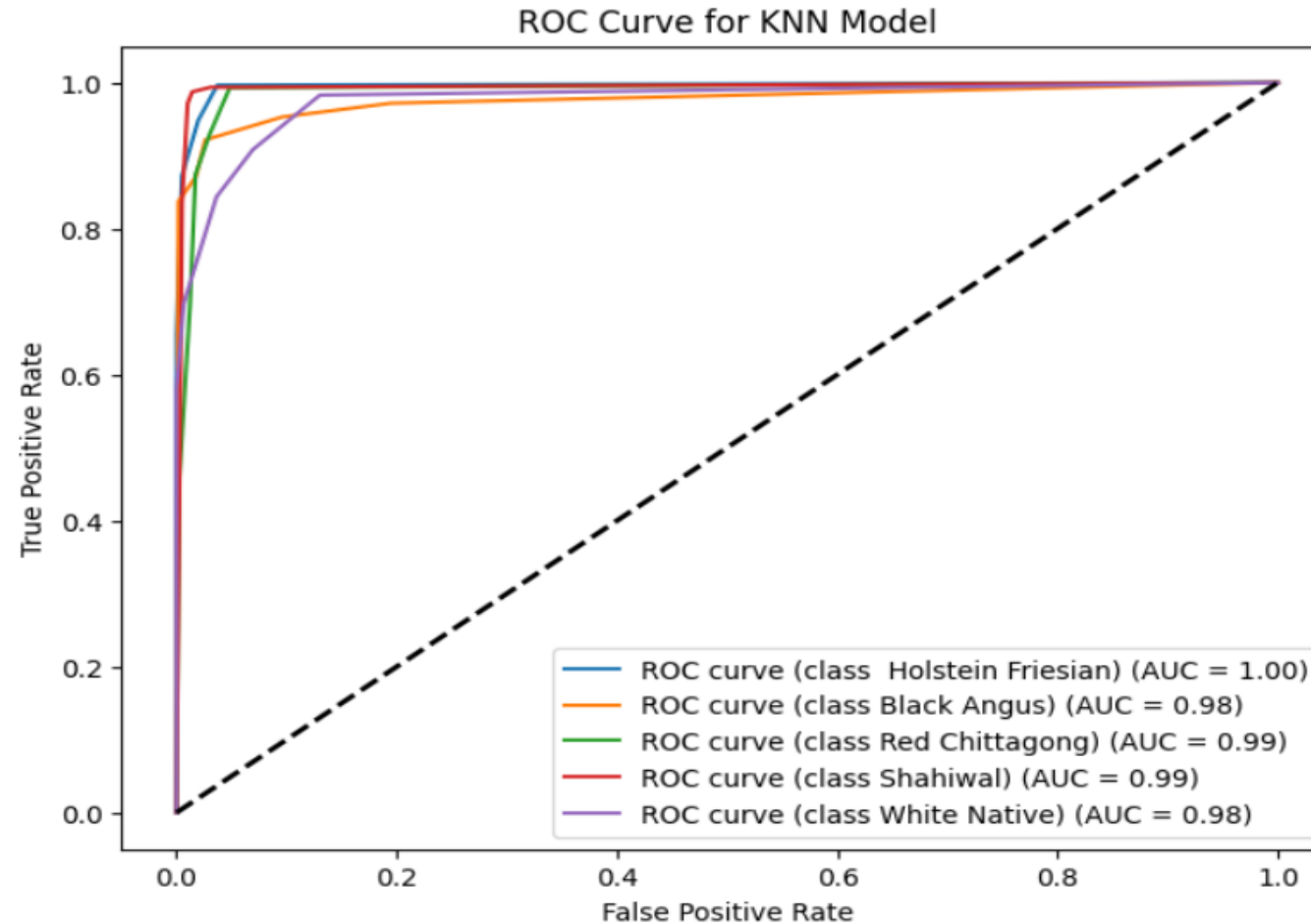


Figure 36: ROC curve of cattle breed recognition



RESULT & OBSERVATION (CONT.)



Measuring performance according to ROC curve for Random Forest model

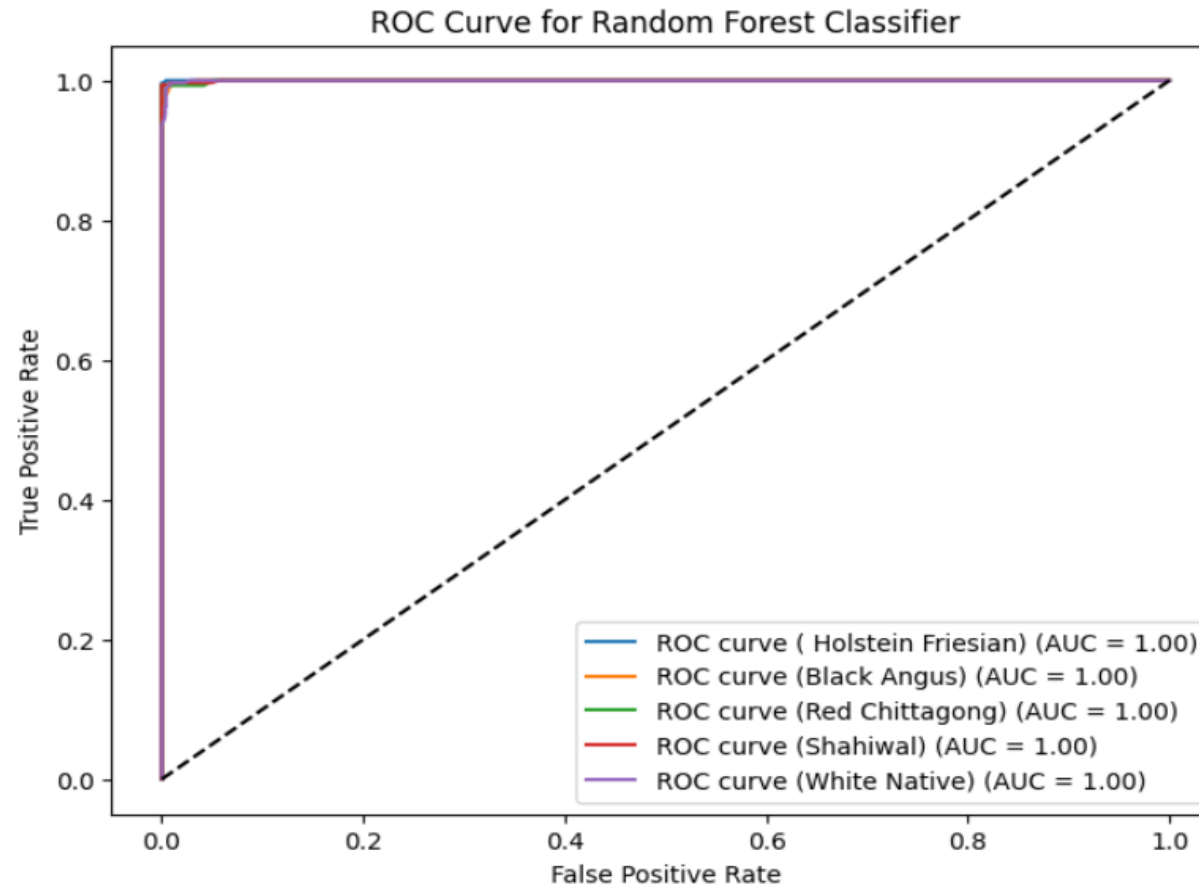


Figure 37: ROC curve of cattle breed recognition



Table 4. Comparative Insights of ROC-AUC Curve

Cattle Breed	CNN	SVM	KNN	Decision Tree	Random Forest
Holstein Friesian	0.97	0.99	1.00	0.96	1.00
Black Angus	0.98	0.99	0.98	0.96	1.00
Red Chittagong	1.00	1.00	0.99	0.99	1.00
Sahiwal	1.00	1.00	0.99	0.99	1.00
White Native	0.96	0.99	0.98	0.95	1.00

Performance Comparison

Comparison with related works on Cattle breeds classification

- Extensive Dataset Variety
- High Accuracy Across Models
- Effective Use of Data Augmentation
- Random Forest Model Excellence



LIMITATION



- ❑ The dataset may not fully encompass global cattle breeds, impacting the model's versatility.
- ❑ Our models are limited to recognizing only five distinct cattle types.

CONCLUSIONS

- ☐ Propose a image scratching model
- ☐ Proper detection of cattle breeds type
- ☐ Accurately classifying the specific cattle breeds showcased in the dataset
- ☐ Empowering farmers
- ☐ Achieves high accuracy



FUTURE RECOMMENDATIONS



- ☐ Continuously update the dataset to encompass a more comprehensive range of cattle variations and environmental conditions.
- ☐ Explore the integration of advanced deep learning architectures to enhance classification accuracy.
- ☐ Explore the potential for transfer learning to leverage pre-trained models for improved performance.

REFERENCES

- [1] R. Kasarda, N. Moravčíková, Gábor Mészáros, Mojca Simčič, and D. Zaborski, “Classification of cattle breeds based on the random forest approach,” *Livestock Science*, vol. 267, pp. 105143–105143, Jan. 2023, doi: <https://doi.org/10.1016/j.livsci.2022.105143>
- [2] S. Li et al, “Individual dairy cow identification based on lightweight convolutional neural network,” *PLOS ONE*, vol. 16, no. 11, p. e0260510, Nov. 2021, doi: <https://doi.org/10.1371/journal.pone.0260510>
- [3] M.E. Hossain et al., .“A systematic review of machine learning techniques for cattle identification: Datasets, methods and Future Directions,” *Artificial Intelligence in Agriculture*, vol. 6,pp. 138–155, 2022. doi:10.1016/j.aiia.2022.09.002



Thank You